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Safety data sheet

BASF Safety data sheet according to Regulation (EC) No. 1907/2006 as amended from time to time.
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Product: **Neopor® F 5300 Plus BMB**
(ID no. 30692534/SDS_GEN_GB/EN)
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SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Neopor® F 5300 Plus BMB

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Expanding-agent containing plastic for the production of foam plastics
Recommended use: for industrial use only, Expanding-agent containing plastic for the production of foam plastics

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY

Contact address:
BASF plc
4th and 5th Floors, 2 Stockport Exchange
Railway Road, Stockport, SK1 3GG
UNITED KINGDOM

Telephone: +44 161 475 3000
E-mail address: product-safety-uk-and-ireland@basf.com

1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to Regulation (EC) No 1272/2008 [CLP]

No need for classification according to GHS criteria for this product.

2.2. Label elements

Globally Harmonized System, EU (GHS)

Hazard Statement:

EUH018	In use may form flammable/explosive vapour-air mixture.

Precautionary Statement:

P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P233	Keep container tightly closed.
P243	Take action to prevent static discharges.
P403 + P235	Store in a well-ventilated place. Keep cool.

The product does not require a hazard warning label in accordance with GHS criteria.

2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

May cause some eye irritation which should cease after removal of the product. In use may form flammable/explosive vapour-air mixture.

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Chemical nature

Preparation based on: polystyrene, propellant, polymeric flame-proofing agent
CAS number 1195978-93-8

Hazardous ingredients (GHS)
according to Regulation (EC) No. 1272/2008

pentane

Content (W/W): >= 3.84 % - <= 4.48 % CAS Number: 109-66-0 EC-Number: 203-692-4 INDEX-Number: 601-006-00-1	Asp. Tox. 1 Flam. Liq. 2 STOT SE 3 (drowsiness and dizziness) Aquatic Chronic 2 H225, H304, H336, H411 EUH066

isopentane

Content (W/W): >= 0.96 % - <= 1.12 % CAS Number: 78-78-4 EC-Number: 201-142-8 INDEX-Number: 601-085-00-2	Asp. Tox. 1 Flam. Liq. 1 STOT SE 3 (drowsiness and dizziness) Aquatic Chronic 2 H224, H304, H336, H411 EUH066

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

No special precautions necessary.

If inhaled:
Keep patient calm, remove to fresh air. If difficulties occur: Seek medical attention.

On skin contact:
Wash affected areas thoroughly with soap and water. If irritation develops, seek medical attention.

On contact with eyes:
In case of contact with the eyes, rinse immediately for at least 15 minutes with plenty of water. If irritation develops, seek medical attention.

On ingestion:
No hazards anticipated. Rinse mouth and then drink 200-300 ml of water. If difficulties occur: Seek medical attention.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms: headache, dizziness, coordination disorder, dazed state, Eye irritation, skin irritation

Hazards: No hazards anticipated.

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media

Suitable extinguishing media:

water spray, foam, dry powder, carbon dioxide

Unsuitable extinguishing media for safety reasons:
water jet

5.2. Special hazards arising from the substance or mixture
Endangering substances: carbon monoxide, Carbon dioxide, styrene, aliphatic hydrocarbons
Advice: The substances/groups of substances mentioned can be released in case of fire.

5.3. Advice for fire-fighters
Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Burns with dense emission of soot. Containers/tanks should be cooled with water spray. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. Flammable concentrations of propellant may accumulate on storage in closed containers. Product will burn on contact with flame or exposure to high temperature.

SECTION 6: Accidental Release Measures

High risk of slipping due to leakage/spillage of product. Shut off or stop source of leak.
Substance/product can form explosive mixture with air.

6.1. Personal precautions, protective equipment and emergency procedures
Sources of ignition should be kept well clear. Ensure adequate ventilation. Note that this gas is heavier than air and can spread along the ground in the direction of the wind. Beware of pits and confined spaces. Use antistatic tools. Vapours are heavy and collect in low areas. Avoid all sources of ignition: heat, sparks, open flame.

6.2. Environmental precautions
Do not allow to enter drains or waterways. Discharge into the environment must be avoided.

6.3. Methods and material for containment and cleaning up
For small amounts: Sweep/shovel up. Pack in tightly closed containers for disposal.
For large amounts: Pick up with vacuum equipment approved for use in hazardous locations. Pack in tightly closed containers for disposal.
Ensure adequate ventilation. Dispose of absorbed material in accordance with regulations. Avoid raising dust.

6.4. Reference to other sections
Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling
Avoid dust formation. Avoid inhalation of dusts. Keep away from sources of ignition - No smoking. Take precautionary measures against static discharges. Product may charge electrostatically: always use earthing leads when transferring from one container to another and earth containers. It is recommended that operators should wear antistatic clothing and footwear. Use antistatic tools.

Ensure an efficient ventilation (at least one air change per hour). Provide good room ventilation even at ground level (vapours are heavier than air). Monitoring of the air in work room necessary.

Protect against moisture. Protect from direct sunlight. Protect against heat. Keep container tightly sealed. Containers under pressure should be opened with care to release pressure. Once container is opened, content should be used as soon as possible. Re-open used containers with caution. Provide good ventilation when handling large quantities. Containers should be opened carefully in well-ventilated areas to avoid static discharge. Sealed containers should be protected against heat as this results in pressure build-up.

Processing machines must be fitted with local exhaust ventilation. Avoid the formation and deposition of dust. During transportation in silo trucks the product is covered with nitrogen, do not climb in. Monitoring of the air in work room necessary. Product should be worked up in closed equipment as far as possible. Protect the container opening with a wire mesh cover.

Protection against fire and explosion:
The product is combustible. Vapours may form ignitable mixture with air. Keep away from heat. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy. Avoid all sources of ignition: heat, sparks, open flame. Containers should be earthed during decanting operations. It is recommended that all conductive parts of the machinery are grounded. All parts of the plant and equipment should be electrically bonded together and grounded. Electrical continuity should be checked at regular intervals. Higher line velocity can increase the build-up of static electric charge. Avoid flammable gas mixtures. Ensure an efficient ventilation (at least one air change per hour). Vapours are heavier than air and may accumulate in low areas and travel a considerable distance up to the source of ignition. Because of danger of explosion, avoid vapours reaching the cellar, sewage water and pits. Empty containers may contain flammable residue.

Temperature class: T3 (Autoignition temperature >200 °C).

7.2. Conditions for safe storage, including any incompatibilities

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Paper/Fibreboard, Stove-lacquer RDL 50, Stove-lacquer R 78433, Carbon steel (Iron), Stainless steel 1.4301 (V2), Stainless steel 1.4361, Stainless steel 1.4401, Stainless steel 1.4439, Stainless steel 1.4539, Stainless steel 1.4541, Stainless steel 1.4571, Stainless steel 1.4306 (V2A), Stainless steel 1.4307, Stainless steel 1.4311, Stainless steel 1.4404, Polyamide (PA)
Further information on storage conditions: Protect against heat. Keep away from sources of ignition - No smoking. Keep only in the original container. Keep container tightly sealed. Protect against moisture. Avoid direct sunlight. Protect containers from physical damage. The authority permits and storage regulations must be observed. Store protected against freezing. Keep tanks under inert gas. Air monitoring should be used to alert any build up of explosive mixtures. Equipment to be installed in an environment with potentially explosive atmospheres should conform to the requirements of ATEX Directive 94/9/EC. Ventilate freight container with open door for 30 minutes before unloading.

7.3. Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

Refer to the current edition of HSE Guidance Note EH40 Occupational Exposure Limits (United Kingdom).

78-78-4: isopentane
TWA value 1,800 mg/m3 ; 600 ppm (WEL/EH 40 (UK))
TWA value 3,000 mg/m3 ; 1,000 ppm (OEL (EU))
indicative
109-66-0: pentane
TWA value 1,800 mg/m3 ; 600 ppm (WEL/EH 40 (UK))
TWA value 3,000 mg/m3 ; 1,000 ppm (OEL (EU))
indicative
7782-42-5: Graphite
TWA value 10 mg/m3 (WEL/EH 40 (UK)), Inhalable dust
TWA value 4 mg/m3 (WEL/EH 40 (UK)), Respirable dust

8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if dusts are formed.

Hand protection:
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Safety glasses

Body protection:
Anti-static protective clothing, safety shoes (f.e. according to EN 20346), antistatic

General safety and hygiene measures
Avoid inhalation of dusts/mists/vapours. No special precautions necessary. When using do not eat or drink. When using do not smoke.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

Form:	beads
Colour:	black
Odour:	of pentane
pH value:	
	not soluble
softening temperature:	approx. 70 °C
Boiling point:	
	not applicable

Sublimation point:
not applicable
Flash point:
Vapours are flammable.
Information on: pentane
Flash point: *-56 °C*

Evaporation rate:
The product is a non-volatile solid.
Flammability: not highly flammable (UN Test N.1 (ready combustible solids))
Lower explosion limit:
Product not examined: Value is calculated from the data of the components.
Information on: pentane
Lower explosion limit:
For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.

Upper explosion limit:
Product not examined: Value is calculated from the data of the components.
Information on: pentane
Upper explosion limit:
For liquids not relevant for classification and labelling.

Ignition temperature: 285 °C (DIN 51794)
Vapour pressure:
not applicable
Density: approx. 1.02 - 1.05 g/cm3
(20 °C)
Relative vapour density (air):2.5
Heavier than air.
Solubility (qualitative) solvent(s): aromatic hydrocarbons, ketones, organic solvents
soluble
Partitioning coefficient n-octanol/water (log Kow):
not applicable
Self ignition: not self-igniting Test type: Spontaneous self-ignition at room-temperature.

Thermal decomposition: approx. 220 °C
No decomposition if used as directed.
Explosion hazard: not explosive
Fire promoting properties: not fire-propagating

9.2. Other information

Minimum ignition energy:	0.28 mJ The product is not capable of a dust explosion.
Bulk density:	approx. 650 kg/m3 (20 °C)
Miscibility with water:	immiscible

SECTION 10: Stability and Reactivity

10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated., Vapours may form explosive mixture with air.

10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

10.3. Possibility of hazardous reactions

Formation of explosive gas/air mixtures.

10.4. Conditions to avoid

> 70 °C
Avoid all sources of ignition: heat, sparks, open flame. Avoid direct sunlight. Avoid electro-static discharge.

10.5. Incompatible materials

Substances to avoid:
explosive substances according UN transport regulations class 1, Propellant release will be boosted with increasing temperature.

10.6. Hazardous decomposition products

Possible thermal decomposition products:
pentane
styrene monomers, Heated product evolves combustible vapours.

SECTION 11: Toxicological Information

11.1. Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
not applicable

Experimental/calculated data:
LD50 (oral): > 2,000 mg/kg

LC50 (by inhalation): > 5 mg/l

LD50 (dermal): > 2,000 mg/kg

Irritation

Assessment of irritating effects:
No irritation is expected under intended use and appropriate handling.

Experimental/calculated data:
Skin corrosion/irritation: non-irritant

Serious eye damage/irritation: non-irritant

Respiratory/Skin sensitization

Assessment of sensitization:
There is no evidence of a skin-sensitizing potential.

Experimental/calculated data:
Non-sensitizing.

Germ cell mutagenicity

Assessment of mutagenicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Carcinogenicity

Assessment of carcinogenicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Specific target organ toxicity (single exposure)

No data available.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Aspiration hazard

No data available.

Other relevant toxicity information

No reports of ill effects provided product was correctly handled and processed.

SECTION 12: Ecological Information

12.1. Toxicity

Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. No toxic effects occur within the range of solubility.

Aquatic invertebrates:
EC50 (48 h) > 100 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
Nominal concentration. The product has low solubility in the test medium. An eluate has been tested. No toxic effects occur within the range of solubility. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic plants:
EC50 (72 h) > 100 mg/l (growth rate), Desmodesmus subspicatus (OECD Guideline 201, static)
Nominal concentration. The product has low solubility in the test medium. An eluate has been tested. No toxic effects occur within the range of solubility. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
On the basis of the data available concerning eliminability/degradation and bioaccumulation potential, longer-term harm to the environment is improbable. No data available concerning biodegradation and elimination.

In accordance with the required stability the product is not readily biodegradable. The product has not been tested. The statement has been derived from the structure of the product. The product is virtually insoluble in water and can thus be separated from water mechanically in suitable effluent treatment plants.

Elimination information:
Non-biodegradable.

12.3. Bioaccumulative potential

Bioaccumulation potential:
The product will not be readily bioavailable due to its consistency and insolubility in water.

12.4. Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: Study scientifically not justified.

12.5. Results of PBT and vPvB assessment

The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative).

12.6. Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

12.7. Additional information

Add. remarks environm. fate & pathway:
Because of the product's consistency and low water solubility, bioavailability is improbable.

Other ecotoxicological advice:
At the present state of knowledge, no negative ecological effects are expected. No toxic effects occur within the range of solubility.

Information on: pentane
Other ecotoxicological advice:
The substance has a very low Global Warming Potential and no Ozone Depleting Potential.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Surplus, unused, old beads may still contain residual pentane. Therefore product has to be treated using all the safety measures in place for the fresh material. See Also Section 7.
Recover or recycle if possible
Disposal is via incineration operated by an accredited disposal contractor.
Dispose of contents in a useful bundle in accordance with local, state or national legislation.

The UK Environmental Protection (Duty of Care) Regulations (EP) and amendments should be noted (United Kingdom).

Contaminated packaging:
Remove all packaging for recovery or disposal

SECTION 14: Transport Information

Land transport

ADR

UN number	UN2211
UN proper shipping name:	POLYMERIC BEADS, EXPANDABLE
Transport hazard class(es):	-
Packing group:	III
Environmental hazards:	no
Special precautions for user:	Tunnel code: D/E Can release flammable vapors. No smoking.Ventilate freight container with open door for at least 30 minutes before unloading.

RID

UN number	UN2211
UN proper shipping name:	POLYMERIC BEADS, EXPANDABLE
Transport hazard class(es):	-
Packing group:	III
Environmental hazards:	no
Special precautions for user:	Can release flammable vapors. No smoking.Ventilate freight container with open door for at least 30 minutes before unloading.

Inland waterway transport

ADN

UN number	UN2211
UN proper shipping name:	POLYMERIC BEADS, EXPANDABLE
Transport hazard class(es):	-
Packing group:	III
Environmental hazards:	no
Special precautions for user:	Can release flammable vapors. No smoking.Ventilate freight container with open door for at least 30 minutes before unloading.

Transport in inland waterway vessel

Not evaluated

Sea transport

IMDG

UN number:	UN 2211
UN proper shipping name:	POLYMERIC BEADS, EXPANDABLE
Transport hazard class(es):	9
Packing group:	III
Environmental hazards:	no Marine pollutant: NO
Special precautions for user:	Can release flammable vapors. No smoking.Ventilate freight container with open door for at least 30 minutes before unloading.

Air transport

IATA/ICAO

UN number:	UN 2211
UN proper shipping name:	POLYMERIC BEADS, EXPANDABLE
Transport hazard class(es):	9
Packing group:	III
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for user:	Can release flammable vapors. No smoking.Ventilate freight container with open door for at least 30 minutes before unloading.

14.1. UN number
See corresponding entries for “UN number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Transport in bulk according to Annex II of MARPOL and the IBC Code

Regulation:	Not evaluated
Shipment approved:	Not evaluated
Pollution name:	Not evaluated
Pollution category:	Not evaluated
Ship Type:	Not evaluated

Further information
This product is subject to the most recent edition of "The Carriage of Dangerous Goods and Use of Transportable Pressure Equipment Regulations" and their amendments (United Kingdom).

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
Listed in above regulation: no

The data should be considered when making any assessment under the Control of Substances Hazardous to Health Regulations (COSHH), and related guidance, for example, 'COSHH Essentials' (United Kingdom).

Directive 94/62/EC on packaging and packaging waste

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

15.2. Chemical Safety Assessment

Chemical Safety Assessment not required

SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

In addition to the information given in the safety data sheet we refer to the product specific 'Technical Information'.

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

Asp. Tox.	Aspiration hazard
Flam. Liq.	Flammable liquids
STOT SE	Specific target organ toxicity — single exposure
Aquatic Chronic	Hazardous to the aquatic environment - chronic
EUH018	In use may form flammable/explosive vapour-air mixture.
H225	Highly flammable liquid and vapour.
H304	May be fatal if swallowed and enters airways.
H336	May cause drowsiness or dizziness.
H411	Toxic to aquatic life with long lasting effects.
H224	Extremely flammable liquid and vapour.
EUH066	Repeated exposure may cause skin dryness or cracking.

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.